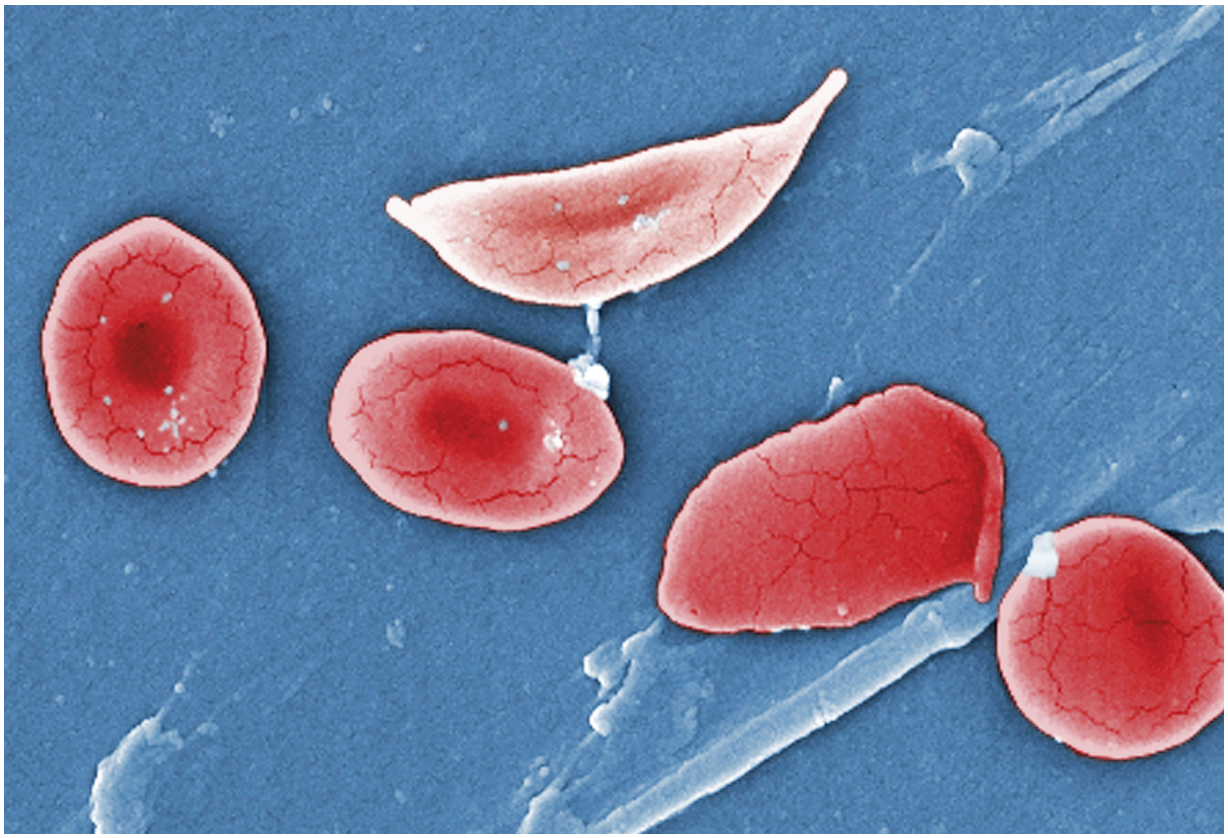


Mutated blood cells occur in sickle-cell disease—a genetic disorder passed on when both parents carry the faulty gene. It can be painful and increases the risk of serious infections.

The human genome

On April 14, 2003, scientists completed the lengthy task of mapping (sequencing) the entire human genome. Geneticists worked out the precise position of all the base pairs in a chain of some three billion of the base nucleotides comprising an estimated 30,000 individual genes. This has allowed geneticists to identify new genes and the role they play in organisms.

Armed with this knowledge, an individual can find out if they have inherited a faulty gene from a parent. Additionally, with access to such data it is possible to screen embryos for known genetic disorders before implantation in the womb. By June 2021, scientists had sequenced the